Statistics and Data Analysis - Lab-03

Distrubution of a statistic and confidence intervals

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Point estimation

Estimation gives a **single** value as an estimate of the population parameter.

Example:

Let $X = (X_1, \ldots, X_n)$ be a sample of people's height

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i = 170$$

But what if we took some other sample? Would the mean differ greatly?

It depends...

- on the distribution
- and the sample size.

Confidence intervals

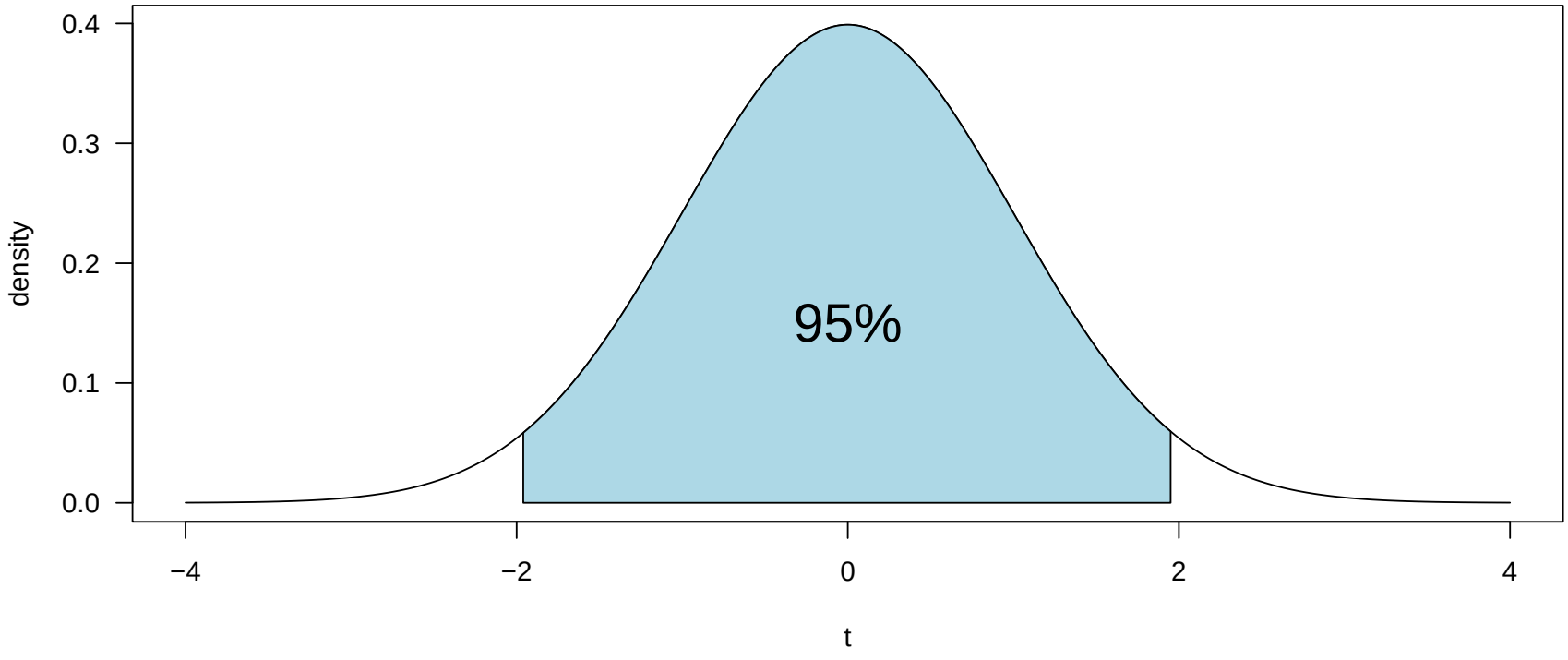
Assume that $X \sim \mathcal{N}(\mu, \sigma)$ and we know σ, then

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \quad \Rightarrow \quad \frac{\bar{X} - \mu}{\sigma}\sqrt{n} \sim \mathcal{N}(0, 1)$$

We can also recall Central Limit Theorem!

So for large samples from distributions satisfying CLT assumptions this will also be approximately true.

Confidence intervals



$$\mathbb{P}\left(\bar{X} - z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

But what if we do not know the distribution of our sample?

Bootstrap sampling

Having just one sample and not knowing the distribution it comes from, we can only estimate a statistic, but not its confidence intervals or variance of this statistic.

Bootstrap sampling is a statistical method used to overcome this problem and estimate the accuracy (e.g., variance, confidence intervals) of a sample statistic.

How to create a bootstrap sample

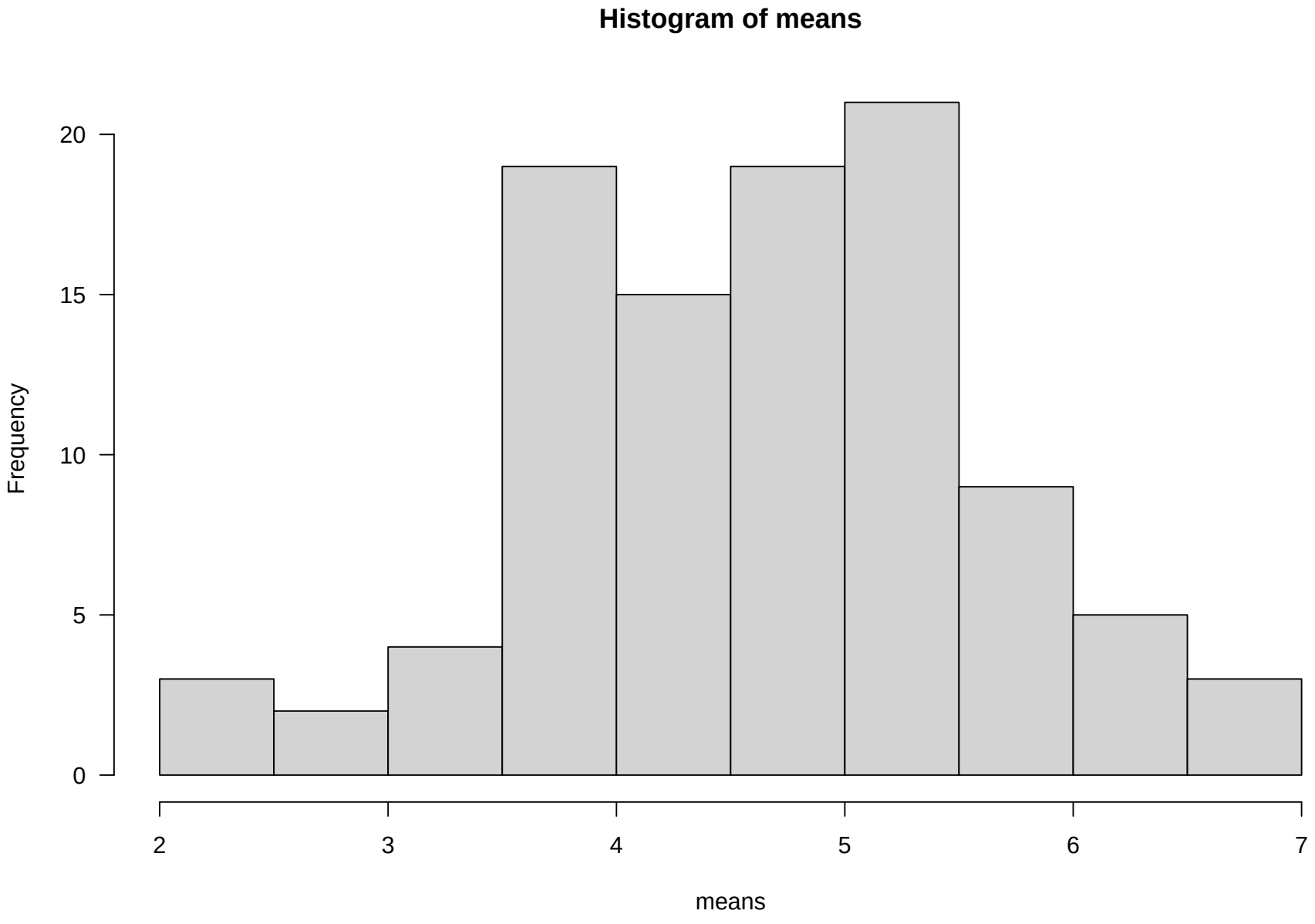
1. Random resampling (with replacement) from an existing dataset.
2. Calculating the statistic of interest for each such sample.
3. Using the distribution of those statistics to estimate uncertainty.

Example

[0, 1, 2, 3, 4, 5, 6, 7, 8, 9] -> 4.50

[1, 1, 2, 0, 0, 9, 6, 6, 9, 6] -> 4.00
[4, 2, 7, 2, 1, 3, 8, 6, 4, 3] -> 4.00
[6, 8, 7, 9, 3, 6, 1, 9, 6, 4] -> 5.90
[0, 9, 5, 8, 2, 6, 9, 6, 6, 7] -> 5.80
[6, 9, 1, 7, 0, 8, 0, 8, 5, 9] -> 5.30
[7, 6, 2, 3, 1, 3, 1, 4, 1, 3] -> 3.10
[3, 2, 0, 9, 3, 4, 7, 6, 2, 6] -> 4.20
[8, 6, 7, 7, 6, 3, 1, 1, 9, 7] -> 5.50
[2, 3, 2, 0, 0, 9, 9, 7, 8, 2] -> 4.20
[5, 8, 1, 6, 5, 0, 5, 8, 0, 5] -> 4.30

Example



Example

(1) Start with the original sample of size $n = 10$. Cretae a list for storing results `m`.

```
x = np.array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
m = []
```

(2) Create a sample of length $n = 10$ from this sample: sampling with replacement

```
y = np.random.choice(x, size=x.shape[0], replace=True)
```

(3) Compute the statistic of interest, e.g a mean on a new sample

```
m.append(y.mean())
```

(4) Reapeat steps (2) and (3) $N = 100$ times $\Rightarrow$ we have a new sample `m` consisting of $N = 100$ means.

(5) Now we can calculate its variance or find confidence intervals by calculating appropriate quantiles of a new sample.

```
np.quantile(m, [0.025, 0.975])
```